Summary Report: Introduction to Supervised Learning Algorithms

Project Overview

This project introduces the basic concepts of Supervised Learning using Python and the Scikit-learn library. Three commonly used supervised learning algorithms were implemented to understand their working principles and applications.

The project includes one regression algorithm and two classification algorithms using beginner-friendly datasets.

Datasets Used

1. Diabetes Dataset

- Source: Scikit-learn Built-in Dataset
- Type: Regression
- Algorithm Used: Linear Regression

2. Wine Dataset

- Source: Scikit-learn Built-in Dataset
- Type: Classification
- Algorithms Used:
  - Logistic Regression
  - Decision Tree Classifier

Algorithms Implemented

Linear Regression

Linear Regression is a supervised learning algorithm used to predict continuous numerical values. In this project, it was implemented using the Diabetes dataset to demonstrate a basic regression model.

Logistic Regression

Logistic Regression is a classification algorithm used to predict categorical outcomes. The Wine dataset was used to classify wine samples into different classes.

Decision Tree

Decision Tree is a supervised learning algorithm that creates a tree-like structure to make predictions. It was implemented using the Wine dataset for classification.

Project Workflow

1. Imported the required Python libraries.
2. Loaded the Diabetes and Wine datasets.
3. Split the datasets into training and testing sets.
4. Trained Linear Regression, Logistic Regression, and Decision Tree models.
5. Generated sample predictions to demonstrate the implementation of each algorithm.

Conclusion

This project provided a beginner-friendly introduction to supervised learning algorithms using Python. It demonstrated the implementation of Linear Regression, Logistic Regression, and Decision Tree models on built-in Scikit-learn datasets. The project helped in understanding the basic workflow of supervised machine learning, including data preparation, model training, and prediction.